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Thermodynamic characteristics of multi-temperature graded compressed air energy storage system

Gen Li^{1,2}, Huan Guo^{2,3*}, Yujie Xu^{2,4,5*} and Haisheng Chen^{2,4,5}

Abstract

Compressed air energy storage (CAES) is one of the effective technologies for addressing the intermittency and variability of solar and wind power. However, the relative narrow power range of conventional CAES hinder efficient integration and stable operation. To address this, a novel Multi-Temperature Graded Compressed Air Energy Storage (MTG-CAES) system is proposed, aiming to broaden the power input/output range, improve energy utilization efficiency, and enhance renewable energy accommodation capability. During charging, the system absorbs wind and solar photovoltaics power by a two-stage adiabatic compressor and a single-stage isothermal compressor, and compression heat are recovered. Meanwhile, wind power with high-frequency fluctuations that is difficult to absorb is converted into heat through resistive heating and stored in the thermal storage system. During discharging, a three-gradient energy release strategy employs the integration of solar thermal energy, compression heat, and surplus wind-generated heat. This enables graded air heating at the expander inlet and tiered power regulation. The results show that the output power range can be extended to 279.17 MW to 682.36 MW and the input power range can be extended to 210.44 MW to 511.32 MW. The system maintains high efficiency across different temperature gradient design conditions, with exergy efficiency and round-trip efficiency reaching 70.47%, and 72.34%, respectively. Increasing the adiabatic compressor pressure ratio improves efficiency but narrows the power range. Higher solar temperatures significantly widen the power range, offering advantages in regions with abundant solar resources. A smaller discharge-to-charge time ratio can also widen the power range. The MTG-CAES system offers new options for the depth and use of renewables.

Keywords Compressed air energy storage, Multi-temperature graded, Wide working range, Efficiency, Renewable energy

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1 Introduction

Renewable energy has experienced rapid development in recent years. According to the International Energy Agency (IEA), its share in the power sector is projected to reach 46% by 2030 [1]. Solar and wind power have now become the second and third largest energy sources in China, respectively. Additionally, wind and solar energy exhibit strong temporal and seasonal complementarity. However, alongside this dramatic increase in installed capacity, the inherent intermittency and instability of wind and solar power have led to increasingly prominent issues of wind and solar curtailment [2–5], particularly

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pronounced in scenarios with low system integration levels and inadequate measures [6]. Energy storage is a key solution to address the intermittency challenges of wind and solar power [7]. Compressed Air Energy Storage (CAES) systems offer significant advantages, including large storage capacity, long storage time, high system efficiency, and extended operational lifetime [8]. Furthermore, CAES can store and convert multiple energy forms, including electricity, heating, demonstrating good compatibility and ease of integration with various energy systems such as solar, wind, and geothermal energy [9]. Wind energy and solar energy are naturally complementary across temporal dimensions covering diurnal and seasonal changes as well as weather conditions [10]. CAES, with its multi-energy conversion capabilities, is exceptionally well-suited to capitalize on this synergy [11]. When integrated with CAES systems, they can form an integrated solution featuring power generation, energy storage and output, which significantly enhances the dispatchability, economic viability and grid friendliness of renewable energy. This combination stands as one of the key technical pathways to drive energy transition and build new-type power systems, and is especially applicable to scenarios with high penetration of renewable energy. Consequently, coupling CAES with solar and wind energy holds significant practical significance. Integration approaches are diverse, primarily involving thermoelectric coupling, optimization of operational strategies, and improvements in system configuration [12–20].

Numerous scholars have conducted in-depth research on wind-solar compressed air energy storage integrated systems. Chen, et al. [15] demonstrated the advantages of CAES simultaneously coupled with both wind and solar energy and validated its technical feasibility. By analyzing China's wind-solar resources, salt cavern distribution, and power grid layout, they identified regions most suitable for wind-solar compressed air energy storage development in China. Research on wind-solar compressed air energy storage integration primarily focuses on two areas: (1) enhancing the intrinsic efficiency of wind-solar compressed air energy storage systems, and (2) multi-objective optimization of system dispatch strategies.

Regarding efficiency enhancement in wind-solar CAES systems, Hammann, et al. [16] reported a 13.2% improvement in system efficiency for wind-Solar CAES compared to solar-only CAES systems. Ji, et al. [17] proposed a novel configuration integrating an Organic Rankine Cycle for graded energy utilization. Results demonstrate that this system achieves an electric energy storage efficiency of 87.7% and an exergy efficiency of 65.4%. Wu, et al. [18] proposed an integrated thermochemical wind-solar compressed air energy storage system that utilizes solar heat

to drive the endothermic reduction of Co_3O_4 to CoO . This heat is subsequently employed during discharge to preheat the expander inlet air, with the proposed system achieving an exergy efficiency of 75.6%. Deymi, et al. [19] developed a wind-solar compressed air energy storage system for near-zero energy buildings that simultaneously supplies cooling, heating, and electricity. The system meets 99% of the building's electrical demand and 61% of its heating load, achieving peak energy efficiency of 41% in December. Zoghi, et al. [20] developed a stable hydrogen production system integrating solar, wind, biomass, and geothermal energy. Four configurations were designed, each prioritizing one renewable source as the baseline while utilizing others as supplementary, all targeting stable hydrogen output. The biomass-based system achieved the highest exergy efficiency of 28.62% but was the only configuration producing CO_2 emissions.

Other researchers focus on dispatch strategies and multi-objective optimization for wind-solar CAES integrated systems. They primarily aim to optimize system efficiency, economic performance, and load-tracking capability, employing algorithms like NSGA-II to determine optimal operating parameters and strategies. Wang, et al. [21] proposed a load-following wind-solar compressed air energy storage system optimized for capacity configuration, targeting balanced power curtailment, load deficiency mitigation, and system investment costs. The optimized system enhances utilization of wind-solar complementarity. Davoodi, et al. [22] developed an off-grid wind-solar compressed air energy storage system for islands/coastal areas, supplying cooling, heating, power, and freshwater. Using energy efficiency, total revenue, and water production as optimization objectives, TOPSIS multi-objective optimization determined the system's optimal pressure ratio. Liu, et al. [23] introduced an improved model-free deep reinforcement learning algorithm, TD3-AC, into an integrated energy system incorporating power grid, wind turbines, photovoltaics, Advanced Adiabatic-CAES (AA-CAES), heat pumps, and adsorption chillers. This approach reduced operational costs by 8.3% while enhancing dispatch accuracy by 29.13%.

Moreover, the intermittent nature of renewable energy sources often forces CAES systems to operate in off-design states, leading to unstable power absorption and delivery. This operational instability is further amplified by fluctuating grid demands, pressure dynamics in storage caverns, and varying ambient conditions. Consequently, numerous researchers have focused on methods to broaden the CAES operating envelope. These approaches fall into two categories: component-level variable-geometry control and system-level operating strategies and configurations. At the component level,

variable-geometry techniques include guide-vane control [24–28], variable-speed operation [29–32], throttle control [33, 34], and nozzle control [35, 36].

In contrast to component-level local optimizations that target individual devices, system-level strategy adjustments and configuration optimizations adopt a holistic perspective, enabling coverage of a broader range of system operating conditions. By coordinating the operating states of all subsystems, they not only expand the operational envelope but also effectively safeguard overall system efficiency, preventing performance losses that can arise from isolated component adjustments. Wang, et al. [37] proposed and validated a matching-analysis method for a multi-compressor series–parallel configuration, built upon variable-speed and variable-guide-vane geometric adjustments. Their results demonstrate that integrating control strategies with series–parallel compressor arrangements can effectively expand the system's operating range. Zhang, et al. [38] proposed a variable-configuration CAES system. Beyond the baseline AA-CAES setup, they added a low-pressure air storage reservoir and installed clutches between the first two and last two stages of both the compressor and expander. By switching between low-power and high-power operating modes, the operating range of the compressor and expander can be extended by up to 70.85% and 27.27%, respectively. Zhang, et al. [39] proposed a CAES system coupled with stepped stage-change thermal storage, noting that increasing the number of adiabatic stages leads to a reduction in the system's output power.

At present, research on coupling renewable energy with CAES has largely concentrated on enhancing system efficiency, optimizing economic performance, and refining dispatch strategies. However, research efforts focused on expanding the operational range of such integrated systems remain limited. Likewise, efforts to broaden the operating range of standalone CAES have predominantly focused on internal component optimization and operational-strategy improvements; investigations into leveraging external topology innovations or synergistic use of external energy sources to extend this envelope are relatively limited. In fact, both the efficient utilization of renewable energy and the stable operation of CAES entail multi-energy storage and conversion processes, and CAES possesses significant technical advantages for long-time, large-scale storage, rendering it indispensable for facilitating renewable energy integration. Therefore, exploring approaches that combine renewable-energy synergy with novel topological configurations to broaden the CAES operating envelope holds considerable theoretical importance and practical value.

Building on the above, we propose a novel multi-temperature graded compressed air energy storage

(MTG-CAES) system. In this configuration, During the charging stage, high-frequency wind power that cannot be directly absorbed is converted into thermal energy and stored in heat tanks, which not only prevents resource waste but also broadens the power absorption range of the energy storage system. During the discharging stage, solar thermal energy, compression heat, and surplus wind-generated heat jointly serve as heat sources. A three-gradient, temperature-graded thermal storage tank array successively recovers the waste heat from each preceding gradient to achieve stepwise regulation of the expander-inlet air temperature, thereby producing three distinct power output levels under the design conditions. Moreover, when combined with conventional control measures—such as flow modulation—the system's operational power range can be substantially broadened. The proposed MTG-CAES not only extends the adjustable power range while maintaining high system efficiency, but also realizes efficient staged energy utilization and markedly enhances renewable-energy accommodation. This paper focuses on the steady-state thermodynamic characteristics of the MTG-CAES system. When studying the impact of different temperature levels and varying wind power heating ratios on system operating conditions, the system's power range and performance can be reflected. In addition, this paper also investigates the influence of key parameters including adiabatic compressor pressure ratio, solar thermal temperature, expander stage number and wind-to-solar heat ratio on the system's thermodynamic performance, so as to provide references for the research on the integration of renewable energy and compressed air energy storage systems.

2 System description

Figure 1a shows the process flow diagram of the MTG-CAES, which operates through distinct charging and discharging stages. Figure 1b shows the flow path of thermal oil. During the charging stage, electricity generated by wind turbines and photovoltaic modules drives two-stages adiabatic compressor followed by a single-stage isothermal compressor to compress air for storage in an air storage tank. The isothermal compressor adopted in this study can be a spray-assisted reciprocating piston compressor [40–42]. As the high-temperature air exhausts from the adiabatic compressors, it is cooled by thermal oil, which recovers the compression heat. Since the temperature of this recovered heat is close to that of the second thermal gradient, the heat is stored in the second gradient thermal tank (G2-HT). The isothermal compressor actively suppresses unnecessary temperature rise through precise thermal control, effectively preventing additional heat generation. The charging stage not only absorbs electrical energy through compressor operation,

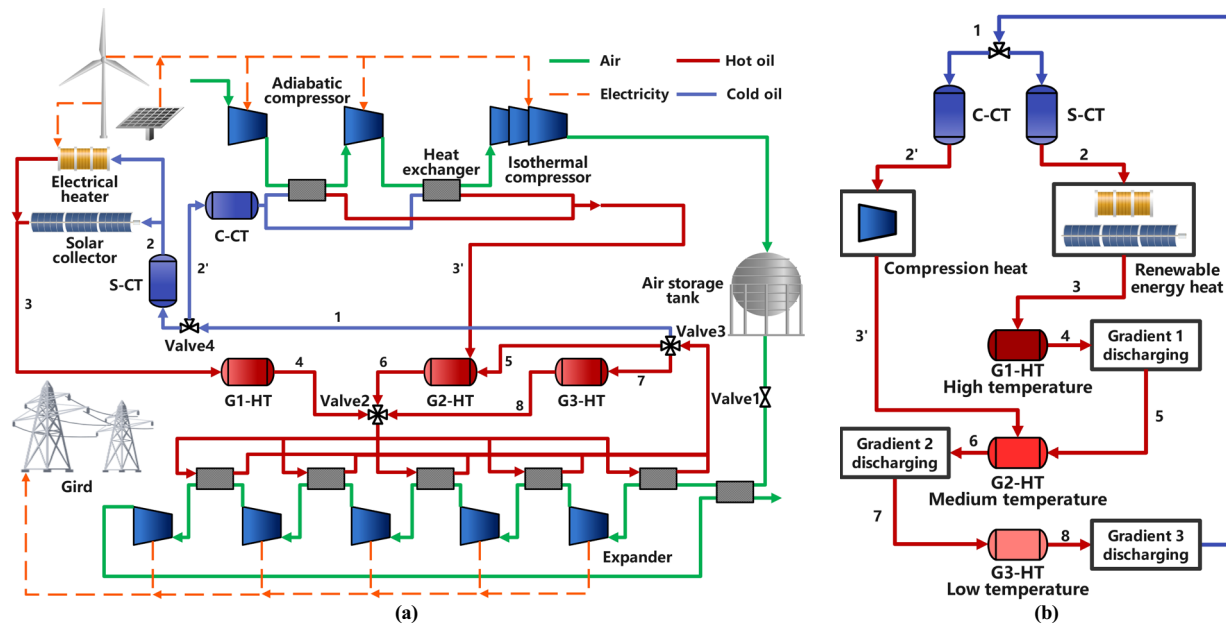


Fig. 1 a Schematic diagram of the MTG-CAES system (b) Flow path of thermal oil (C-HT: Compression heat tank; C-CT: Compression cold tank; S-CT: Solar cold tank; G1-HT: First gradient heat tank; G2-HT: Second gradient thermal tank; G3-HT: Third gradient thermal tank)

but also stores thermal energy via solar collectors and electrical heaters, which respectively capture solar thermal energy and high-frequency wind power that cannot be utilized by the compressors. During periods of limited solar availability—such as nights, autumn/winter seasons, or cloudy days when wind resources are typically abundant—surplus wind power is converted to heat via electrical heater. This supplements potential solar thermal deficits anticipated during the discharging stage.

During the charging stage, the system increases operational flexibility by dynamically allocating incoming renewable energy among air compression, solar thermal collection, and wind-to-heat conversion via electric heaters. By exploiting the temporal complementarity of wind and solar resources and using multiple forms of energy storage, this approach smooths charging-power fluctuations, improves accommodation of variable renewables, and builds stable, well-conditioned energy reserves that enable subsequent flexible and efficient discharging.

During the discharging stage, the system utilizes thermal oil from three gradient heat tanks to heat and expand high-pressure air through a tiered temperature approach for power generation. The discharging stage under the three temperature gradients are as follows:

- (1) The first gradient operation: High-pressure air released from the storage tank is first throttled down to 7 MPa. This air then undergoes preheating in a recuperator that captures waste heat from the last-stage expander's exhaust. Subsequently,

the preheated air enters a heat exchanger where it is further heated by high-temperature oil from the solar-charged G1-HT before expanding through the first expander. Crucially, each expander inlet is equipped with interstage heaters, all sourcing heat exclusively from G1-HT to maximize work output. After power generation, the discharged oil at 244°C is channeled into the G2-HT and mixed with compression heat at a temperature similar to that of the discharged oil.

- (2) The second gradient operation follows an identical flow path but uses blended thermal oil from G2-HT as its heat source. After power generation, the discharged oil at 168°C is channeled into the G3-HT.
- (3) Finally, the third thermal gradient utilizes the hot oil recovered from the second gradient, and the used hot oil is fed back to S-CT and C-CT in the original proportion.

Flexible switching among different temperature gradients can be achieved through coordinated control of Valve 2, which regulates the routing of thermal oil from different gradient thermal tanks into the heat exchanger, thereby adjusting the inlet air temperature of the expander. The return of thermal oil is managed by controlling Valve 3, allowing the discharged thermal oil to be recovered into the corresponding thermal tanks. The system process is described in a gradient sequence above to

clearly and intuitively illustrate the conceptual operating logic, rather than to imply a strictly fixed operating order. In practical operation, the selection of temperature gradients during the energy release process is highly flexible and can be dynamically adjusted according to variable grid load demands, the temperature and thermal storage states of each thermal tank, and the availability of renewable energy input, thereby enabling accurate adaptation to complex and varying load scenarios.

This proposed configuration establishes three well-defined temperature gradients, directly enabling three distinct power output levels for grid flexibility. The system's core advancement lies in its staged energy utilization: During charging, it combines adiabatic compression with isothermal compression while integrating solar thermal storage and wind-powered thermal storage; during discharging, it creates a thermal hierarchy by integrating solar heat with recovered compression heat. By sequentially heating expander inlet air with these graded heat sources, the system achieves multi-level power generation. Significantly, the first gradient's direct use of solar heat dramatically enhances renewable energy integration capacity, positioning MTG-CAES as an adaptable solution for modern power grids.

3 Methodology

The proposed system was modeled in MATLAB software with fluid properties retrieved via REFPROP. The simulation incorporates the following assumptions:

- (1) Heat exchanger pressure drops are considered, while pipeline pressure losses are neglected.
- (2) Thermal dissipation losses from air and fluids are ignored.
- (3) Air leakage across all components is disregarded.

Several idealized assumptions are introduced, as pipeline pressure losses have been reported to have a negligible impact on system performance [43], and temperature variations caused by thermal dissipation have only a minor effect on efficiency [26]. The influence of air leakage is generally small [44]. Most previous studies have similarly neglected these effects [45–48]. Therefore, the results represent an upper-bound performance under ideal operating conditions, serving as a theoretical benchmark rather than direct practical performance.

3.1 Component modeling

The adiabatic work consumption of compressor is expressed as:

$$W_{AC} = M_s(h_{AC,2s} - h_{AC,1})/\eta_{AC} \quad (1)$$

where W_{AC} is the work consumption of adiabatic compressor, M_s is stored air mass, $h_{AC,1}$ is inlet specific enthalpies. $h_{AC,2s}$ is the outlet specific enthalpy for an ideal isentropic process. η_{AC} is the isentropic efficiency of compressor.

The isothermal work consumption compressor is expressed as:

$$W_{IC} = M_s R_g T_{IC,in} \ln(P_{IC,out}/P_{IC,in})/\eta_{iso} \quad (2)$$

where W_{IC} is work consumption of the isothermal compressor; R_g is gas constant of air; $T_{IC,in}$ is inlet temperature of the isothermal compressor; $P_{IC,in}$ and $P_{IC,out}$ are inlet and outlet pressures of the isothermal compressor respectively; η_{iso} is isothermal efficiency.

The output work of the expander is expressed as:

$$W_T = M_r(h_{T,1} - h_{T,2s}) \cdot \eta_T \quad (3)$$

where W_T is the output work of expander; M_r is air mass during the discharging stage; $h_{T,1}$ is enthalpy at expander. $h_{T,2s}$ is the outlet specific enthalpy for an ideal adiabatic process. η_T is the isentropic efficiency of expander.

The heat transfer in the heat exchanger is expressed as:

$$Q_{HE} = M_{cold}\Delta h_{cold} = M_{hot}\Delta h_{hot} \quad (4)$$

where Q_{HE} is heat transfer rate of the heat exchanger; M_{cold} and M_{hot} are mass flow rates of cold and hot streams in the heat exchanger respectively; Δh_{cold} and Δh_{hot} are enthalpy changes of cold and hot streams respectively. The heat exchange temperature difference is $\Delta T = 5^\circ\text{C}$.

This study calculates the exergy destruction distribution within the proposed system. By analyzing the relative magnitudes of exergy destruction across components, key factors affecting system exergy efficiency are identified, providing clear guidance for system optimization. The calculation methods for component-level exergy destruction are as follows:

The exergy destruction of compressor is given by:

$$E_{s,C} = E_{C,in} + W_C - E_{C,out} \quad (5)$$

where $E_{s,C}$ is the exergy destruction of the compressor, $E_{C,in}$ and $E_{C,out}$ represent the exergy of air at the compressor inlet and outlet, respectively, and W_C denotes the work input to the compressor.

The exergy destruction of the expander is expressed as:

$$E_{s,T} = E_{T,in} + W_T - E_{T,out} \quad (6)$$

where $E_{s,T}$ is the exergy destruction of the expander, $E_{T,in}$ and $E_{T,out}$ are the exergy of air at the expander inlet and outlet, respectively, and W_T represents the work output of the expander.

The exergy destruction in the heat exchanger is defined as:

$$E_{s,HE} = E_{cold,in} - E_{cold,out} + E_{hot,in} - E_{hot,out} \quad (7)$$

where $E_{s,HE}$ denotes the exergy destruction of the heat exchanger, $E_{cold,in}$ and $E_{cold,out}$ are the exergy values of the cold stream at the inlet and outlet, and $E_{hot,in}$ and $E_{hot,out}$ represent the exergy of the hot stream at the inlet and outlet.

The exergy destruction due to throttling in the air storage tank is calculated as:

$$E_{s,TANK} = E_{air,in} - E_{air,out} \quad (8)$$

where $E_{s,TANK}$ is the exergy destruction of the air storage tank, and $E_{air,in}$ and $E_{air,out}$ are the exergy of air at the tank inlet and outlet, respectively.

The exergy of the working fluid, referenced in the above equations, is determined by:

$$E_X = (H - H_0) - T_0(S - S_0) \quad (9)$$

where E_X is the exergy of the working fluid, H and S are its specific enthalpy and entropy, respectively, and the subscript 0 denotes properties at the ambient reference state.

3.2 Evaluation indicators

Three thermodynamic evaluation metrics are employed here, namely system exergy efficiency, energy efficiency, and round-trip efficiency accounting for solar energy converted work [49], which are specified as follows:

The system exergy efficiency is

$$\eta_{exergy} = \frac{W_T}{W_C + E_{re}} \quad (10)$$

where η_{exergy} is the system exergy efficiency; W_C is the total work consumption of compressors, including both the adiabatic and isothermal compressors; W_T is the total work output of expanders, including the expanders summed over all gradients; E_{re} is the input exergy of solar thermal energy and wind-powered electric heating. E_{re} is shown:

$$E_{re} = E_{solar} + E_{wind} \quad (11)$$

where E_{solar} is solar input exergy, E_{wind} is input exergy of wind-powered electric heating:

$$E_{solar} = Q_{solar} \left(1 - \frac{T_0}{T_s}\right) \quad (12)$$

$$E_{wind} = \frac{Q_{wind}}{\eta_{eh}} \quad (13)$$

where Q_{solar} is the solar thermal input heat; T_0 is the ambient temperature; T_s is defined as the nominal outlet temperature under design operating conditions and is treated as a constant value. In Sect. 4.2, the thermodynamic performance of the proposed system under different T_s values is further investigated. Q_{wind} is the heat from wind-powered electric heating; and η_{eh} is the electric heating efficiency.

Then, the system energy efficiency is defined as:

$$\eta_{energy} = \frac{W_T}{W_C + Q_{re}} \quad (14)$$

where η_{energy} is the system energy efficiency; Q_{re} is the heat required by the system.

The system round-trip efficiency is defined as follows [50, 51]. Since solar and wind heating are external auxiliary energy only for enhancing discharge work, their equivalent output work is deducted to accurately characterize the core energy conversion efficiency of the system:

$$\eta_{roundtrip} = \frac{W_T - W_{wind} - W_{solar}}{W_C} \quad (15)$$

where $\eta_{roundtrip}$ is the system round-trip efficiency accounting for solar thermal converted work. W_{solar} is the solar thermal converted work, expressed as follow.

$$W_{solar} = \frac{Q_{solar} \eta_{solar}}{\eta_{collector}} \quad (16)$$

where η_{solar} is the solar thermal power generation efficiency, taken as 0.15; $\eta_{collector}$ is the solar collector efficiency, taken as 0.6667 [51].

To explore system performance at different temperature levels, the gradient exergy efficiency is defined as:

$$\eta_{exergy,Gn} = \frac{W_{T,Gn}}{k_{M,Gn}(W_C + E_{re})} \quad (17)$$

where $\eta_{exergy,Gn}$ is the exergy efficiency of the n th gradient; $W_{T,Gn}$ is the work output of the expander at the n th gradient; $k_{M,Gn}$ is the mass proportion of air at the n th gradient; the subscript G_n denotes the n th gradient.

3.3 Calculation process

Based on above equations, Fig. 2 shows the calculation flow of the MTG-CAES system. The computational procedure is divided into the charging stage and discharging stage. During the charging stage, initial design parameters are input, starting with a given initial air mass flow rate. In the adiabatic compression stage, air properties at the adiabatic compressor inlet are retrieved via REFPROP. Based on the isentropic efficiency and Eq. (1), the outlet parameters are calculated and serve as the inlet conditions for the subsequent heat exchanger. A

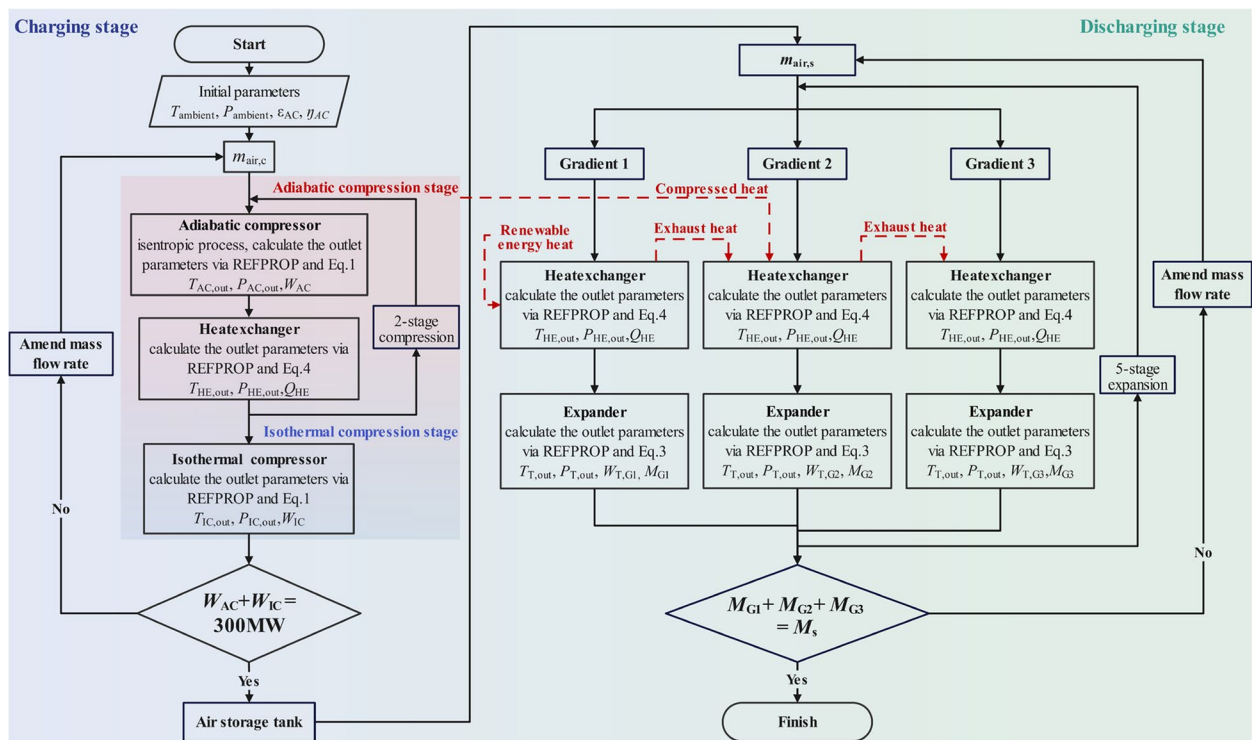


Fig. 2 Calculation logic of the MTG-CAES System

fixed 5°C terminal temperature difference is set in the heat exchanger, and the outlet parameters are computed using REFPROP and Eq. (4), which then become the inlet parameters for the next adiabatic compressor. The sequence of one adiabatic compressor paired with one heat exchanger constitutes a single calculation loop. Since the proposed system adopts two-stage adiabatic compression, this loop is executed twice, with the outlet parameters from the previous stage serving as the inlet for the next. The final outlet parameters from the adiabatic compression stage become the inlet for the isothermal compression stage. Here, REFPROP is again called, and based on the constant-temperature characteristic and Eq. (2), the outlet parameters of the isothermal compressor are determined. The system targets a total charging power of 300 MW. Therefore, after completing the compression calculations, a power verification step checks whether the sum of the adiabatic and isothermal compression power meets 300 MW. If not, the air mass flow rate is adjusted and the entire charging stage calculation is repeated. Once the power criterion is satisfied, the final outlet parameters are recorded as the inlet conditions for the air storage tank. The pressure ratios of all compressor stages are fixed, resulting in a final storage pressure of 10 MPa. The discharging stage comprises calculations for three separate gradients. Starting from the stored air

parameters, a throttling process reduces the pressure to 7 MPa, followed by cooling to ambient temperature. An initial discharging air mass flow rate is assumed. Each gradient utilizes its specific heat source to warm the air in a heat exchanger before it enters the expander model. Within the expander model, inlet properties are obtained via REFPROP, and the outlet parameters are computed according to the isentropic process and Eq. (3). These outlet parameters then become the inlet for the next heat exchanger in the sequence. The proposed system configuration includes five expanders; thus, this expansion cycle repeats five times. The condition for stopping the discharging process is that the discharge flow rate makes the sum of air masses in the three gradients equal to the air mass stored in the air storage tank during the charging stage, meaning all the stored air is fully expanded. If this condition is not met, the discharge air flow rate is adjusted.

4 Results and discussion

Basic parameters of the system are shown in Table 1. The charging time is treated as a fixed value of 8 h, selected to represent a typical large-scale daily energy storage scenario [52, 53]. The discharging time is determined by the total amount of stored compressed air

Table 1 Basic parameter of system

Parameter	Value	Parameter	Value
Ambient temperature	25 °C	Discharging pressure	7 MPa
Ambient pressure	0.1 MPa	Expansion ratio of expander	2.1925
Charging air flow rate	512.22 kg/s	Number of expander stages	5
Charging power	300 MW	Expander isentropic efficiency	0.88
Number of adiabatic stages compressor	2	Discharging air mass flow rate	1024.44 kg/s
Pressure ratio of adiabatic compressor	6.1032	Total air mass in air storage tank	14751947 kg
Pressure ratio of isothermal compressor	2.7532	Grade1 discharging power	568.63 MW
Adiabatic compressor isentropic efficiency	0.84	Grade2 discharging power	484.45 MW
Isothermal compressor efficiency	0.9	Grade3 discharging power	398.82 MW
Temperature of compression heat	300°C	Grade1 discharging capacity	140.87 MWh
Storage pressure	10 MPa	Grade2 discharging capacity	959.27 MWh
Solar thermal power generation efficiency	0.15	Grade3 discharging capacity	706.84 MWh
Solar collector efficiency	0.6667	Heat exchanger pressure drop	0.02 MPa
Charging time	8 h	Heat exchanger temperature difference	5 °C
Discharging time	4 h	Solar temperature	350 °C

during the charging process and the air mass flow rate during discharging. Under the baseline parameters listed in Table 1, the corresponding discharging time is 4 h.

We investigated the effects of adiabatic compressor pressure ratio, solar temperature, expander stage number, and discharge/charge time ratio on gradient output power, as well as the impact of wind-to-solar heat ratio on system performance, as these parameters represent the primary factors governing system efficiency and input/output power range.

4.1 Analysis of typical operating conditions

The system performance under baseline parameters is shown in Fig. 3. The proposed system achieves an

exergy efficiency of up to 70.47%, an energy efficiency of 66.56%, and a round-trip efficiency of 72.34%. For Gradient 1 to 3, the exergy efficiencies are 88.70%, 75.58%, and 62.21% in sequence; the designed output powers are 568.63 MW, 484.50 MW, and 398.82 MW in sequence; and the discharge capacities are 140.87 MWh, 959.27 MWh, and 706.84 MWh in sequence. The designed input power range can reach 300 MW to 450.63 MW. When integrated with a flow regulation range of 0.7 to 1.2, the output power range can reach 279.17 MW to 682.36 MW, and the input power range can reach 210.44 MW to 511.32 MW. The proposed system can achieve 79.95% and 83.19% of input and output power with 375 MW and 484.5 MW as the baseline for input and output power, respectively. The sequential decrease in both

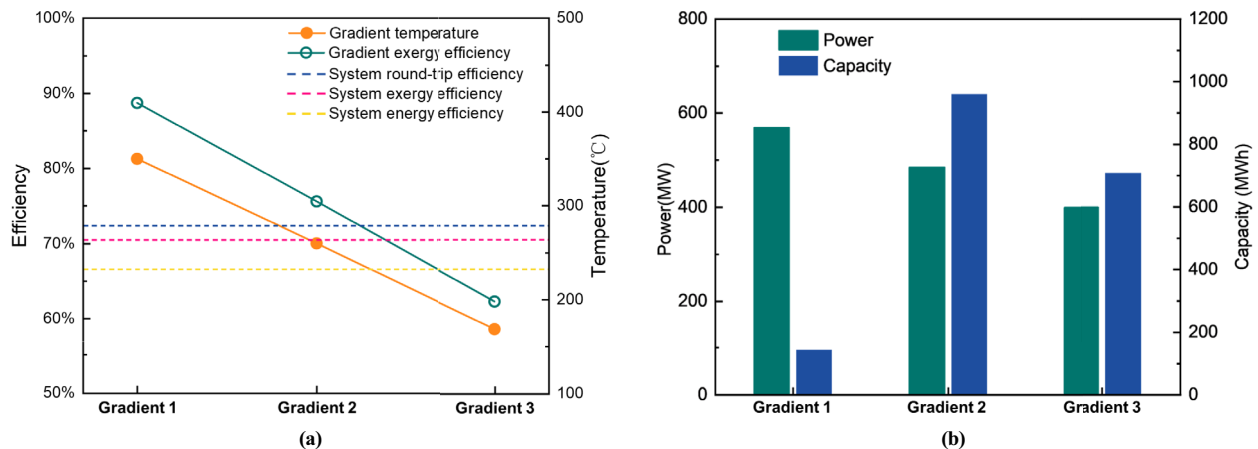


Fig. 3 The system performance under baseline parameters. **a** Gradient temperature and exergy efficiency; System efficiencies. **b** Gradient power and capacity

gradient exergy efficiency and output power across stages is attributed to the progressive reduction in heat source temperature at each gradient. The higher gradients, with their elevated heat source temperatures, enable superior heating of the pressurized air, resulting in greater power output during expansion and consequently higher exergy efficiency. The capacity distribution across gradients is primarily governed by the corresponding air mass allocation. The distribution of air mass among gradients is further governed by both the mass and temperature of their respective heat sources, as these two parameters collectively determine the amount of air that can be effectively heated within each operational stage.

Table 2 presents a literature comparison of the power output range between the MTG-CAES system and other types of improved CAES systems. The calculation method for the output power range is defined as the ratio of the difference between the maximum and minimum output power to the nominal output power. It can

be seen that the efficiency of the MTG-CAES system is higher than that of all systems reported in the referenced literature. In terms of output power range, only the system with a multi-pressure reservoirs structure proposed by Zhang [38] exhibits a larger power output range than the MTG-CAES system; however, its system efficiency is significantly lower than that of the MTG-CAES system. Therefore, the multi-temperature gradients structure proposed in this work demonstrates superior performance. Specific comparisons are shown in Table 2:

Figure 4 shows the exergy destruction distribution within the MTG-CAES system. The analysis reveals that the compression-expansion unit group and the heat exchange subsystem account for the most significant exergy destructions, representing 16.21% and 8.03% of the total input exergy, respectively. Additionally, the air tank throttle valve contributes considerable exergy destruction.

Compared to conventional AA-CAES systems, the proposed system employs a higher pressure ratio in the

Table 2 Comparison between different systems

Ref	Characteristic	Nominal equipment isentropic efficiency		System efficiency		Output power range
		Compressor	Expander	Exergy efficiency	Roundtrip efficiency	
This system	Multi-temperature gradients	84%	88%	70.47%	72.34%	83.19%
Guo [25]	Variable stator vane	84%	88%	-	71.83%	62.80%
Guo [54]	Sliding-pressure operation	84%	88%	-	71.83%	53.21%
Baigorri [55]	High-temperature air-based concentrated solar power plant	84%	88%	-	43.10%	70.45%
Yan [30]	Variable speed	89%	93%	-	65.2%	71.93%
Zhang [38]	Multi-pressure reservoirs	84%	88%	-	60.79%	93.60%
Zhao [56]	Variable inlet pressure	88%	80%	45.16%	-	70.20%

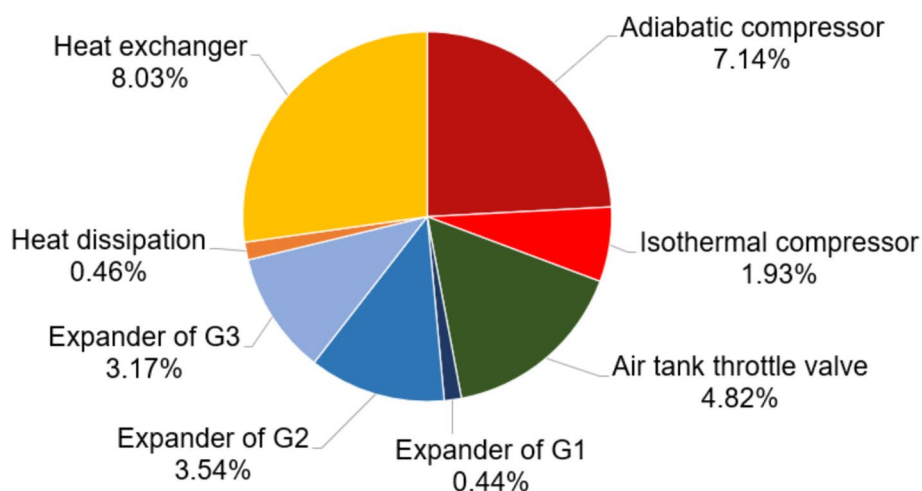


Fig. 4 Exergy destruction proportion of each component relative to the total input exergy

adiabatic compressor to achieve elevated compression heat temperatures. This design choice actually results in reduced exergy destruction [57]. Simultaneously, the incorporation of an isothermal compressor effectively minimizes additional compression heat generation, thereby yielding lower exergy destructions.

During the discharging stage, the exergy destruction across the three gradient expanders follows this descending order: Gradient 2 > Gradient 3 > Gradient 1. This hierarchy primarily stems from the varying air mass distribution among the gradients. As demonstrated in Fig. 5b, the proportion of air mass allocated to each gradient directly correlates with their respective exergy destruction patterns.

The throttling loss across the air tank throttle valve remains substantial. Heat dissipation losses primarily originate from the discharged air across all gradients and the exhausted thermal oil from Gradient 3 during discharging. Their relatively minor contribution can be attributed to two key design features: First, we implemented heat recovery from the final discharged air stream, utilizing it to preheat the inlet air of the primary expander while cooling the exhaust to near-ambient temperature. Second, the three-temperature-gradient utilization strategy for the thermal oil ensures its final discharge temperature remains comparatively low.

4.2 Analysis of system parameters

This section examines the influence of key parameters on system performance, including the effects of adiabatic compressor pressure ratio, solar temperature, expander stage number, and discharge/charge time ratio on gradient output power, as well as the impact of wind-to-solar heat ratio on system efficiency and charging power.

4.2.1 Impact of adiabatic compressor pressure ratio

The adiabatic compressor pressure ratio is a crucial design parameter for this system. It directly affects the outlet air temperature of the adiabatic compressor, the work consumption during the compression stage, and the mass flow rate of the thermal oil that captures compression heat, thereby influencing the overall system performance.

As shown in Fig. 5a, as the adiabatic compressor pressure ratio increases, the outlet air temperature of the adiabatic compressor rises. This results in a higher temperature of the compression heat. With the discharge oil temperature from the first gradient heat tank G1-HT remaining constant, the temperature of the thermal oil in the second gradient heat tank G2-HT increases. This subsequently raises the temperature of the oil in the third gradient heat tank G3-HT, which recovers waste heat from the second gradient.

Furthermore, a higher adiabatic compression ratio leads to an increased mass of the compression thermal oil. Given that the mass of the solar oil remains unchanged, as shown in Fig. 5b, the ratio of compression heat oil to solar thermal oil (oil mass ratio) increases. Since the air mass allocation for each gradient matches the thermal oil mass distribution in their respective heat tanks, this higher oil mass ratio means that the proportion of air mass allocated to the first gradient—where solar heat directly heats the expander inlet air—decreases. Conversely, the air mass shares for the second and third gradients increase. This leads to a reduced demand for solar thermal energy and a slight decrease in renewable energy accommodation capacity.

As illustrated in Fig. 5c, as the pressure ratio of adiabatic compressor increases, under the combined effect of rising heat tank temperatures and increased air mass, both the absolute output work and its proportion for the second and third gradients show an upward trend. Although the temperature of the first gradient heat tank remains stable, its allocated air mass decreases (see Fig. 5b), resulting in reduced output work and a smaller proportion of total work. Under constant total charging power and fixed charging/discharging time, the total stored air mass decreases as the adiabatic compressor pressure ratio increases. This reduces both the charging and discharging air flow rates. As seen in Fig. 5d, the system's duration of the first gradients is significantly shorter than others, which aligns well with the operational characteristics of grid peak shaving where high-load conditions typically persist for brief durations. This time allocation among different gradients mainly results from the selected system configuration, namely the two-stage adiabatic compression and five-stage expansion configuration. In such a configuration, the total heat demand during the expansion stages is primarily covered by the compression heat produced in the charging phase, with only a small portion supplemented by solar and wind energy. As the high-temperature heat input from solar and wind acts as the heat source for the first gradient, only a limited amount of air mass can be assigned to it. Since the energy release flow rate is identical among the three gradients, the operating duration of the first gradient is therefore significantly shorter than that of the other two gradients. The interaction between air mass allocation and discharging flow rates causes the discharging time of the first gradient to shorten, while the discharging time for the second and third gradients lengthen. Furthermore, the work consumption in the adiabatic compression stage increases with the rising adiabatic compressor pressure ratio. Given the fixed total compression power and charging duration as design constraints, the

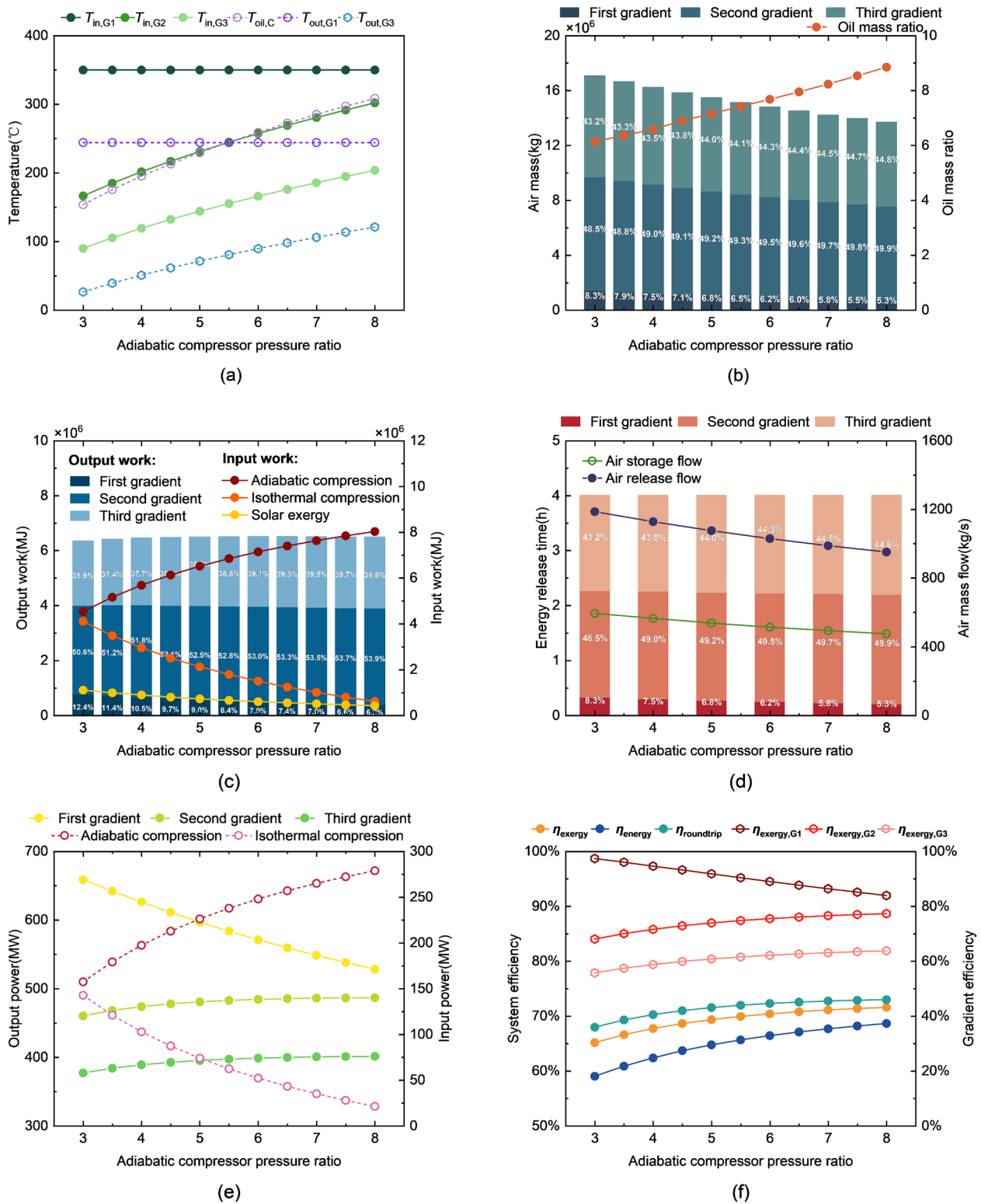


Fig. 5 Impact of adiabatic compressor pressure ratio on system parameters: **a** Temperature of gradient heat tanks, **b** Air mass and oil mass ratio, **c** Output and input work **(d)** Mass flow rates and gradient discharging time, **e** Output and input power, **f** System and gradient efficiency

pressure ratio and work consumption of the isothermal compression stage decrease accordingly.

Figure 5e shows how system power characteristics change with the adiabatic compressor pressure ratio. Similar to the work distribution during charging stage, a higher adiabatic compressor pressure ratio increases the power consumption of the adiabatic compression stage but decreases that of isothermal compression. Thus, while maintaining a constant total charging power of 300 MW, the power trends of these two compression stages are opposite. Additionally, the increase in the pressure ratio of the adiabatic compressor leads to a higher oil mass in the second temperature gradient, which increases the output work of the second and third gradients while correspondingly reducing the output work of the first gradient. Although the discharging time of the first gradient decreases, its output power declines overall due to the significant reduction in output work. In contrast, the discharging times of the second and third gradients change only slightly, and therefore their output power increases. These changes collectively narrow the system's design power output range. Excessively high pressure ratios pose certain challenges for compressor design and manufacturing, while also leading to higher costs. Therefore, the adiabatic compressor pressure ratio should not be set too high.

Figure 5f shows changes in efficiency with increasing pressure ratio. η_{exergy} , η_{energy} , $\eta_{\text{roundtrip}}$, $\eta_{\text{exergy,G2}}$, and $\eta_{\text{exergy,G3}}$ all increase because higher adiabatic compressor pressure ratios elevate the oil temperatures in the second and third gradient heat tanks. This boosts the output work of the second and third gradients, thereby increasing total output work. Meanwhile, total charging work remains constant, and solar thermal demand decreases. However, $\eta_{\text{exergy,G1}}$ decreases because the sharp reduction in air mass allocated to the first gradient substantially lowers its output work. The reduced exergy from pressurized air and solar heat cannot compensate for the efficiency loss caused by this work reduction. Thus, while a higher adiabatic compressor pressure ratio generally enhances overall efficiency, the marginal gains exhibit diminishing returns.

It is important to note that while higher adiabatic compressor pressure ratios improve system efficiency, they simultaneously narrow the power operating range. Therefore, selecting an appropriate pressure ratio requires carefully balancing system efficiency against power output flexibility.

4.2.2 Impact of solar temperature

Unlike the adiabatic compressor pressure ratio which primarily affects the temperature of second and third gradient heat tanks, variations in solar temperature influence

all three gradient heat tanks. The solar temperature corresponds to the inlet oil temperature of the first gradient heat tank G1-HT. As shown in Fig. 6a, an increase in this temperature raises the outlet oil temperature ($T_{\text{out,G1}}$) of the first gradient. When mixed with compression heat oil, this elevates the inlet oil temperature ($T_{\text{in,G2}}$) of the second gradient heat tank G2-HT, which in turn increases the temperature of the third gradient heat tank G3-HT through waste heat recovery. However, only the first gradient heat tank experiences a significant temperature rise, while the temperature increases in the remaining two gradients are relatively small. This occurs because the high oil mass ratio (compression heat to solar thermal oil) minimizes the solar thermal contribution, resulting in limited temperature impact on the second and third gradients.

Figure 6b demonstrates that rising temperatures across all three gradient heat tanks increase work output for each gradient, thereby boosting total output work. The most pronounced work increase occurs in the first gradient due to its substantial temperature rise, leading to a higher proportion of total output work. By contrast, the second and third gradients show smaller work improvements and reduced work proportions. Higher solar thermal temperatures increase the exergy input from solar sources, they do not affect the work and power distribution during the compression stage. This is mainly because the increase in solar thermal temperature directly raises the temperature of the first gradient thermal tank, which increases the inlet temperature of the first gradient expander and enables it to produce more output work. In contrast, the second and third gradients experience only limited temperature increases, and therefore their output work is enhanced to a much smaller extent.

With discharging time remaining unchanged, work and temperature variations directly mirror power output changes. As depicted in Fig. 6c, the first gradient exhibits the most significant power growth due to its dramatic temperature and work increase. The second and third gradients show modest power gains, which closely follow the variation trends of temperature and work output. This dynamic substantially raises the upper limit of discharging power while maintaining the lower limit, resulting in significant expansion of the operational power range under design conditions. However, thermal oil is not suitable for prolonged high-temperature operation, making appropriate and stable solar thermal conditions crucial for the reliable operation of the heat exchange system.

Figure 6d illustrates efficiency responses to solar temperature. η_{exergy} increases with higher temperatures, while both η_{energy} and $\eta_{\text{roundtrip}}$ decline. This divergence stems from distinct treatments of solar inputs: For η_{exergy} ,

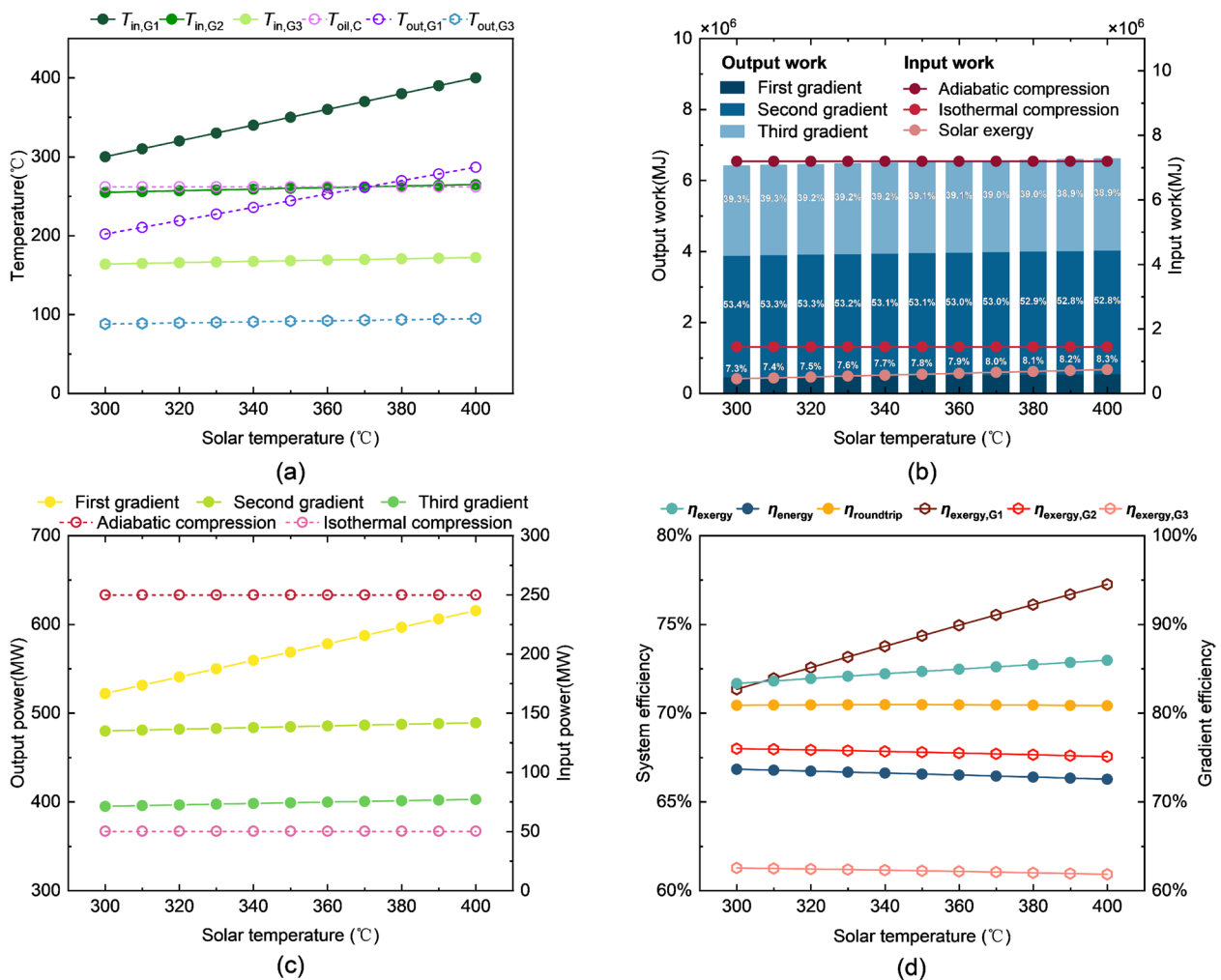


Fig. 6 Impact of solar temperature on system parameters. **a** Temperature of gradient heat tanks, **b** Output and input work, **c** Output and input power, **d** System and gradient efficiency

the work output increase outweighs additional exergy input losses, yielding net efficiency gain. For η_{energy} and $\eta_{roundtrip}$, work gains cannot compensate for increased solar energy input and solar equivalent work conversion losses, causing efficiency reductions. Regarding gradient exergy efficiencies: $\eta_{exergy,G1}$ increases notably due to significant work output growth without proportional increases in pressurized air exergy input or solar exergy share. Conversely, $\eta_{exergy,G2}$ and $\eta_{exergy,G3}$ decrease marginally. Though solar exergy contribution remains small, the lesser temperature increases in these gradients produce insufficient work gains to offset efficiency losses from added solar exergy input.

In summary, increasing the solar temperature enhances the system's operational power range but incurs certain efficiency losses. Moreover, excessively high solar temperatures may compromise the stability of the heat exchange system.

4.2.3 Impact of expander stage number

Changes in expander stage count primarily affect the temperatures of the second and third gradient heat tanks and the mass ratio between compression heat oil and solar thermal oil, subsequently influencing other system parameters. Increasing the number of expander stages reduces the design expansion ratio per stage, raising the outlet air temperature. As shown in Fig. 7a, this elevates the discharge oil temperature from each gradient, thereby increasing temperatures in the second and third gradient heat tanks. This also raises the final exhaust oil temperature. Such temperature variations induce a series of changes in other system parameters.

According to CAES charge–discharge symmetry [58], while recovered compression heat remains constant, more expander stages increase thermal demand during discharging. This deficit must be compensated by additional solar thermal input. Consequently, as seen

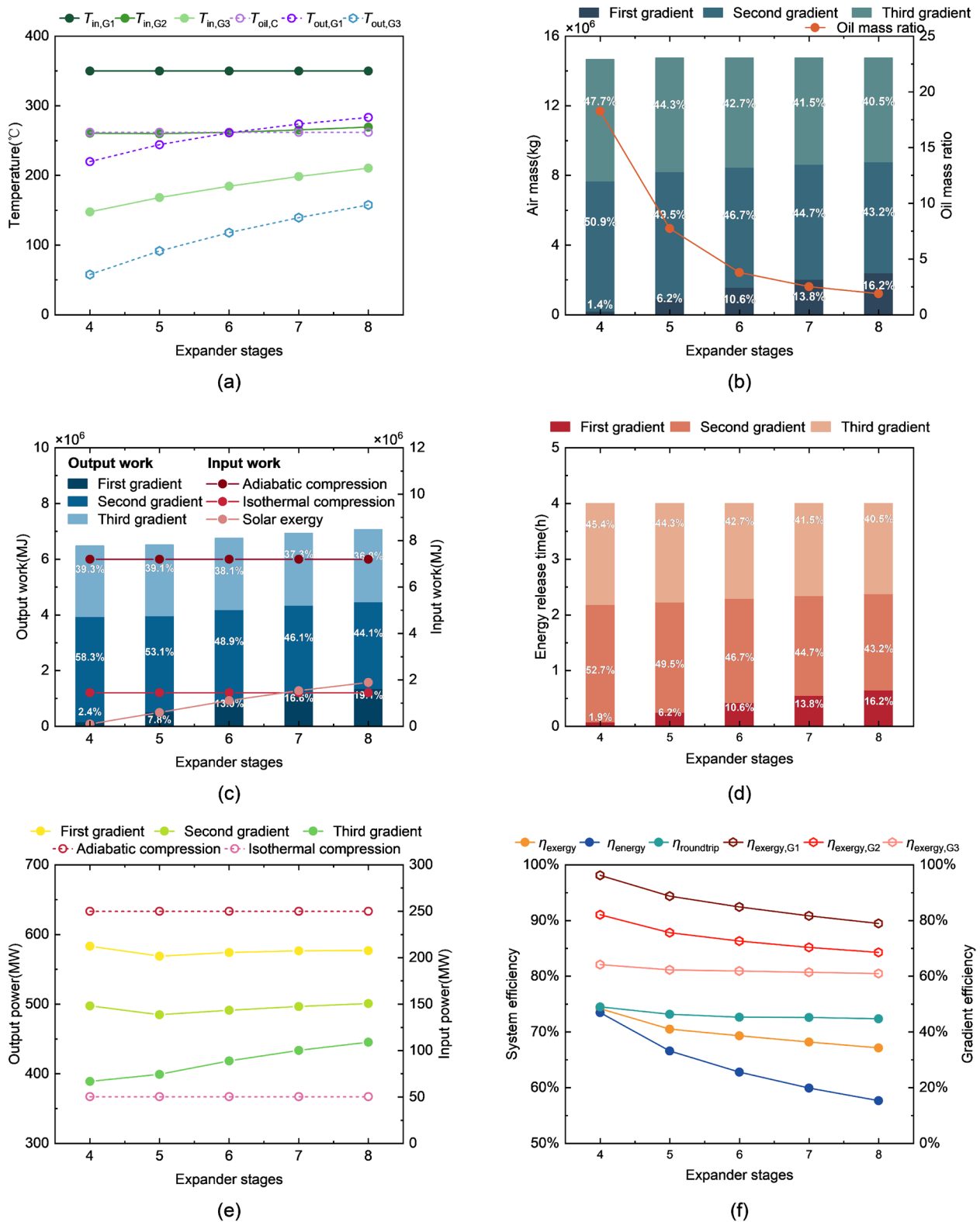


Fig. 7 Impact of number of expander stages on system parameters: **a** Temperature of gradient heat tanks, **b** Air mass and oil mass ratio, **c** Output and input work (**d**) Mass flow rates and gradient discharging time, **e** Output and input power, **f** System and gradient efficiency

in Fig. 7b, the oil mass ratio decreases—meaning more solar thermal oil is allocated to the first gradient. Since air mass distribution corresponds to thermal oil allocation, the first gradient receives increased air mass (Fig. 7c), boosting its work output. Although second and third gradient oil temperatures rise, their reduced air shares slightly decrease work output. With total discharging time fixed at 4 h, discharging time distribution follows air mass allocation changes (Fig. 7d): first gradient discharging extends while second and third gradients shorten. This implies a prolonged duration of high-power operation, indicating that systems with more expander stages can be effectively applied in scenarios requiring sustained high-power output.

Figure 7e shows power variations with expander stages. First and second gradient power exhibit non-monotonic trends—initially decreasing then increasing—due to competing effects of work output and time changes. The first gradient reaches a local maximum as increased work output counteracts extended operation time. The second gradient develops an extremum from reduced work output combined with shorter time. Meanwhile, the third gradient shows distinct power growth despite its lower thermal source temperature because significant time reduction outweighs gradual work increase. These dynamics raise the minimum discharging power, narrowing the overall operating range. Figure 7f illustrates declining efficiency trends with more expander stages, where all system efficiencies (η_{exergy} , η_{energy} , $\eta_{\text{roundtrip}}$) and gradient-level exergy efficiencies decrease for two reasons: An increase in the number of expander stages raises the required solar-thermal input; however, the additional output work produced by this extra solar input is insufficient to compensate for the increased energy input. Moreover, adding expander causes greater exergy destruction across components, so the overall system efficiency declines.

In summary, while a greater number of expander stages extends the duration of high-power gradient operation, it narrows the output power range and increases component exergy destructions, thereby reducing system efficiency. Additionally, employing more expander units raises investment costs. Thus, the selection of the number of expander stages should be based on practical engineering requirements.

4.2.4 Impact of discharging/charging time ratio on gradient output power

Figure 8 shows the variation of output power of each gradient of the system with the discharging/charging time ratio. It can be seen that as the discharging/charging time ratio increases, the energy discharge power of

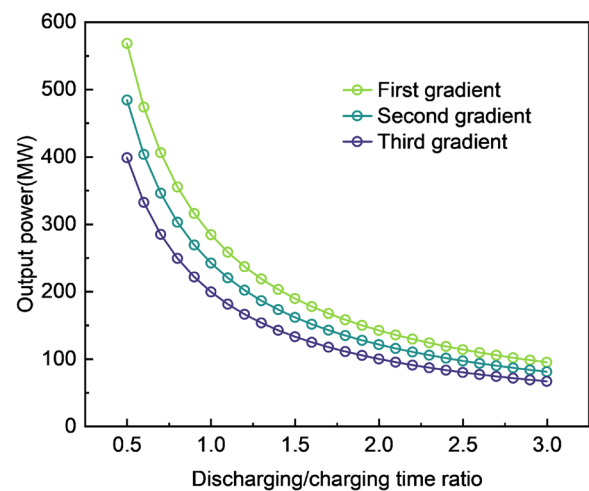


Fig. 8 Variations in gradient output power with discharge/charge time ratio

each gradient shows a decreasing trend, and the decreasing trend gradually becomes gentle. This is because when the discharging time is relatively prolonged, the air flow rate for energy discharge will gradually decrease, which in turn leads to a gradual reduction in energy discharge power.

In addition, the range of energy discharge power gradually narrows as the discharging/charging time ratio increases. This is because the gap between the energy discharge flow rates of the three gradients gradually shrinks with the increase of the discharging/charging time ratio, resulting in a gradual reduction in the gap between the output powers of the three gradients, which in turn narrows the range of energy discharge power.

This indicates that a shorter discharging time or a longer charging time brings a wider power output range to the system. The system further enhances this operational advantage through its integrated utilization of complementary renewable sources. By simultaneously drawing power from both photovoltaic modules and wind turbines during the charging stage, it capitalizes on their natural generation patterns—solar power typically peaks during daylight hours while wind resources often strengthen during nighttime and seasonal transitions. This complementary characteristic enables the system to maintain longer effective charging durations within each operational cycle compared to single-source systems, consequently accumulating more stored energy that directly translates to expanded power output flexibility and range during the discharging stage.

4.2.5 Impact of heat ratio of wind to solar on system efficiency and energy storage power

During solar-insufficient periods (nights, autumn/winter, cloudy days) when wind resources are typically abundant,

wind-powered electric heating supplements solar thermal deficits during discharging, broadening the charging power range. Figure 9 shows the impact of the heat ratio of wind to solar energy on system efficiency and energy storage power. This ratio denotes the proportion of wind-generated heating to solar thermal energy, with energy storage power encompassing both compressor consumption and wind-powered heating. Energy storage power increases at a decelerating rate with higher heat ratio of wind to solar, while system round-trip and exergy efficiencies exhibit declining trends. Additionally, the shorter the time that excess wind power is used for heating, the higher the energy storage power will be. As the heat ratio of wind to solar increases, the exergy efficiency and the round-trip efficiency exhibit different trends. From the perspective of energy conversion pathways during the discharging stage, wind energy and solar energy are utilized through fundamentally different routes. Solar energy follows a “thermal-to-electric” pathway, whereas wind energy follows an “electric–thermal–electric” pathway, which introduces additional irreversible losses. Consequently, as the heat ratio of wind to solar increases, both the exergy efficiency and the round-trip efficiency show an overall decreasing trend due to the higher irreversibility associated with wind energy utilization. The differing decline trends of the two efficiencies mainly arise from the distinct ways in which wind and solar energy are accounted for in their definitions and their positions in the corresponding equations. In the calculation of exergy efficiency, wind and solar inputs appear in the denominator and are evaluated in terms of their exergy values. In contrast, in the round-trip efficiency, these inputs appear in the numerator and are expressed as equivalent work. Owing to these different formulations, the sensitivities of exergy efficiency and round-trip

efficiency to changes in the heat ratio of wind to solar differ, leading to the observed divergence in their trends.

The time proportion of wind-powered electric heating refers to the ratio between the duration of utilizing otherwise unabsorbable high-frequency wind power for heating and the total charging time during the charging process. It should be noted that the period of wind-powered heating does not conflict with conventional charging operations; rather, these two processes occur simultaneously. By adjusting the time proportion of wind-powered electric heating, the power range during the charging stage is expanded from 300 MW to 450.63 MW. When the system operates under off-design conditions, combined with a flow regulation range of 0.7 to 1.2, this range can be further extended to 210.44 MW to 511.32 MW. Although utilizing surplus wind power for heating reduces energy quality and system efficiency, it enhances renewable energy absorption and generates greater economic benefits. Therefore, achieving an optimal balance between charging power absorption and system efficiency is essential.

4.2.6 Summary of sensitivity analysis

The analysis demonstrates that key system parameters exhibit coupled effects, requiring careful trade-offs to balance efficiency and power flexibility. The temperatures at key locations represent critical determinants of both power range and efficiency in the proposed system. Variations in parameters such as adiabatic compressor pressure ratio, solar temperature, and expander stage number directly affect the temperatures across the three gradients during the discharging stage, which in turn govern their respective power output levels. A higher adiabatic compressor pressure ratio improves overall efficiency but narrows the power output range. Increasing the solar temperature expands the operational power range, yet it may reduce energy and round-trip efficiency, with stable conditions being essential for heat exchange reliability. Employing more expander stages extends the duration of high-power discharge but raises exergy destruction, thereby lowering efficiency. A smaller discharge-to-charge time ratio widens the power output range, an effect further enhanced by wind-solar complementary charging. Additionally, a higher wind-to-solar heat ratio significantly boosts charging power and renewable energy absorption, albeit with a slight reduction in system efficiency.

Moreover, although the proposed MTG-CAES system introduces additional components and operational complexity compared with conventional CAES configurations, its economic viability should be assessed by balancing increased capital investment against enhanced operational benefits. On the cost side, the main additional

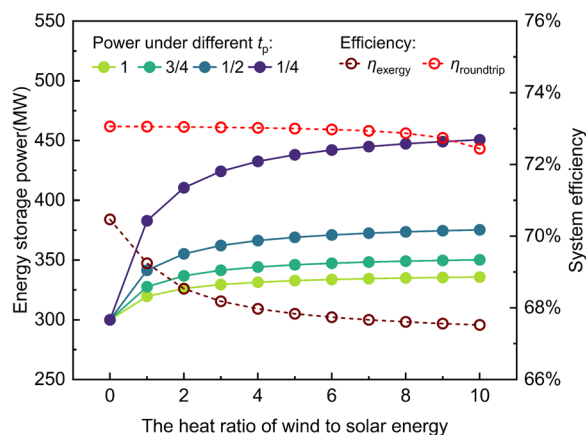


Fig. 9 Impact of heat ratio of wind to solar on system efficiency and energy storage power (t_p : the time proportion of electric heating to total charging)

expenditures arise from the use of multiple thermal energy storage tanks operating at different temperature levels, together with the associated heat transfer equipment and more complex control systems. For systems of similar scale, a single thermal storage tank typically accounts for approximately 1% of the total equipment cost [52]; therefore, the additional storage tanks in the MTG-CAES configuration are expected to increase the initial equipment investment by roughly 3%. On the benefit side, the MTG-CAES system enables multi-level temperature utilization and graded power output, which significantly broadens the effective power regulation range and enhances operational flexibility under variable renewable energy inputs. This expanded power range allows the system to provide more effective peak shaving and load-following services, potentially increasing operational revenue through participation in ancillary service markets. Moreover, enhanced flexibility contributes to reduced renewable energy curtailment and improved utilization of surplus wind and solar electricity. It should be noted that operational revenue is strongly influenced by market mechanisms and policy frameworks, and recent policy developments increasingly favor energy storage deployment, further improving the revenue potential of flexible systems such as MTG-CAES [59–61].

5 Conclusions

This study investigates the integration of wind and solar power with CAES system, with the aim of broadening the system output power range and improving energy utilization efficiency. A novel MTG-CAES system is proposed. A comprehensive thermodynamic analysis is conducted, and system performance is evaluated in terms of exergy efficiency, round-trip efficiency and power range. The results provide references for the integration of renewable energy and CAES systems. The main conclusions are summarized as follows:

- (1) Under the baseline design parameters, the designed output power of the three gradients is 568.63 MW, 484.45 MW and 398.82 MW. The designed input power range can reach 300 MW to 450.63 MW. When combined with a flow regulation range of 0.7 to 1.2, the output power range can reach 279.17 MW to 682.36 MW and the input power range can reach 210.44 MW to 511.32 MW. The system achieves an exergy efficiency of 70.47%, and a round-trip efficiency of 72.34%. The exergy efficiencies of the three heat-release gradients are 88.70%, 75.58%, and 62.21%, and their corresponding discharge capacities are 140.87 MWh, 959.27 MWh, and 706.84 MWh. The above-reported efficiencies are theoretical upper bounds under idealized assumptions.

- (2) The influence of key parameters on the energy discharge stage is revealed. The adiabatic compressor pressure ratio significantly impacts power range and efficiency: Higher ratios improve heat storage temperature and boost Gradient1–3 power output, but reduce total air mass and Gradient1 air allocation, consequently narrowing the power operating range. Solar temperature critically enhances power flexibility: Elevated temperatures substantially raise Gradient1 power capacity, expanding the overall operating range. Although system exergy and energy efficiencies slightly decrease, the round-trip efficiency improves. Increasing expander stages reduces efficiency while substantially raising solar exergy demand and altering gradient power distribution. Higher discharge/charge time ratios contract the power range, indicating that shorter discharging or longer charging time better maintain wide operating flexibility.
- (3) The influence of key parameters on the energy charge stage is revealed. Both increasing the wind-heating proportion and shortening wind-heating absorption time expand the charging power operating range. By adjusting the time proportion of wind-powered electric heating, the power range during the charging stage is expanded from 300 MW to 450.63 MW. When combined with a flow regulation range of 0.7 to 1.2, this range can be further extended to 210.44 MW to 511.32 MW. However, higher wind-heating proportions reduce system efficiency.

The proposed system still faces certain technical challenges. The integration of additional components leads to an increase in initial investment costs, which needs to be offset by the extra operational benefits derived from the advantages of the system. Meanwhile, the flexible switching between different gradients imposes higher requirements on the control system and regulation strategy. In future studies, the economic performance and dynamic characteristics including transient behavior and coordinated control strategies of the MTG-CAES system will be focused on.

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Authors' contributions

Gen Li: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. Huan Guo: Validation, Methodology, Investigation, Formal analysis, Funding acquisition, Conceptualization. Yujie Xu: Validation, Investigation, Writing – review & editing, Supervision. Haisheng Chen: Writing – review & editing, Supervision, Funding acquisition, Project administration.

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Data availability

Data will be made available on request.

Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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